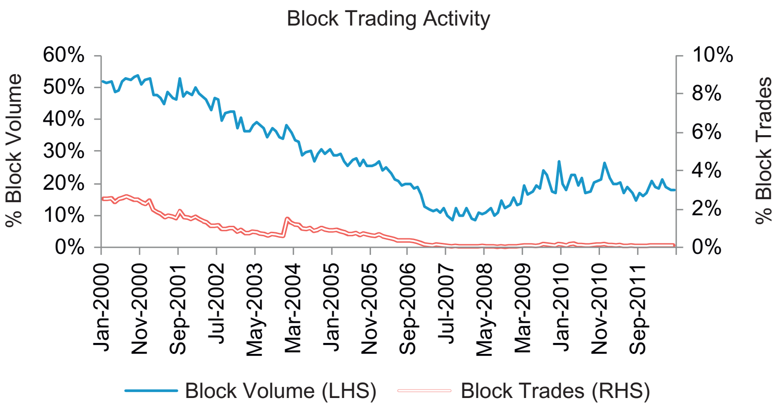

■ **Figure 1.7** Block Trading Activity.

in 2012. The number of block transactions has decreased 95% from about 2.6% of total trades to 0.1% of total trades in 2012. Clearly, trading has become more difficult, measured from the number of transactions required to fill an order.

Our analysis period has seen a dramatic surge in program trading activity over 2000 through 2007. During this time, program trading increased from about 10% of total volume to almost 40%. Then in August 2007, there was a market correction that is believed to have been due to a high correlation across different quantitative strategies and which caused many quantitative funds to incur losses and lose assets under management. Subsequently, program trading activity declined through about September 2008. Moreover, due to the financial crisis many funds turned to program trading to facilitate their trading needs from a risk management perspective, which caused program trading to again increase through 2010 where it seems to have since leveled off at about 30%−35% of total market volume (Figure 1.8).

It is easy to see that the increase in program trading has also helped open the door for algorithmic trading. Program trading is defined as trading a basket of 15 or more stocks with a total market value of $1 million or more. Since 2000, program trading on the NYSE has increased 273% from 10% in 2000 to 36% in 2012. This more than threefold increase in program trading has been attributable to many factors. For example, institutional investors have shifted towards to embracing quantitative investment models (e.g., mean-variance optimization, long-short strategies, minimal tracking error portfolios, stock ranking model, etc.) where the model results are a list of stocks and corresponding share quantities, compared to the more traditional fundamental analysis that only recommends whether a stock should be